PROJECT Session 4 : UCI Adult - Analyzing Income Factors

Deliverables
Step 1: Data Cleaning and Preparation
Step 2 : Feature Transformation and Engineering
Step 3 : Unsupervised Learning (Pattern Discovery)
Step 4 : Visualization and Exploration (with `ggplot2`)
Step 5 : Modeling and Prediction (Data Science)
Questions (answers will be put in the report or the notebook, as you want) :

Groups of 3 people

▼ Deliverables

- **Google Colab Notebook Link :** Provide a link with "Viewer" access. Ensure the notebook is fully executed with all outputs visible.
- **A Brief Report (PDF or included in the notebook) :** A concise summary of the steps taken, your analyses (focusing on the results, not the technical operations), and your conclusions + answers to the questions ; don't forget to put the family names of your group.

It would be appreciated if you use a Notion/Git or any project tool ; that will give you bonus points.

Project Goal : Your team is tasked with analyzing census data to identify the factors that predict whether an individual earns over $50K per year. The main challenge in this dataset is handling complex categorical data and "hidden" missing values. Your objective is to rigorously prepare the data using **PySpark**, consult the dataset's metadata to understand feature definitions, and build both unsupervised and supervised machine learning models.

Source : https://www.kaggle.com/datasets/uciml/adult-census-income

Step 1: Data Cleaning and Preparation

- Be careful when importing the data by taking into account " `?` "character (as a NA value ?)
- `workclass` and `occupation` columns **:** Handle the missing values using the most frequent value of each column.
- **Check the redundancy of the `education` and `education-num` columns :** analyse the redundacy and apply the appropriate manipulations.
- **Categorical variables :** Some categorical variables may contain leading or trailing whitespace. Clean these columns.

Step 2 : Feature Transformation and Engineering

- **Target Transformation:** The `income` variable is text-based. Convert it into a binary variable (0 for `<=50K` and 1 for `>50K`).
- **Create New Features:** Instead of just using the raw data, **create new, meaningful variables**. Think about how you can simplify or enrich the model's understanding. Examples of what you might explore include:
  - Grouping sparse categorical variables (e.g., native countries) into broader categories.
  - Discretizing continuous variables into logical bins (e.g., creating age groups).
  - Combining financial features (e.g., calculating net capital from gains and losses) to see if they are better predictors.

Step 3 : Unsupervised Learning (Pattern Discovery)

Before building a predictive model, explore the dataset's underlying structure using PySpark's MLlib.

- **Algorithm Choice:** Apply either **Principal Component Analysis (PCA)** or **K-Means Clustering** on the continuous variables (like age, hours-per-week, net_capital, etc.).
  - *If PCA:* Reduce the dimensionality and explain the variance captured by the first two principal components. Do they help separate the income classes?

- *If K-Means:* Create 3 to 5 clusters. Analyze the cluster centers—do these natural groupings map to specific income brackets or age groups?

Step 4 : Visualization and Exploration (with `ggplot2`)

Note: Since PySpark does not have native plotting, aggregate the data or sample a fraction of it to convert to a Pandas DataFrame, then use Matplotlib, Seaborn, or Plotly ; you can also extract the data out to do a dashboard on PowerBI, Tableau or other BI tool.

- **Visualization 1 (Categorical vs. Income):** Visualize the proportion of people earning over $50K across different education levels using a bar diagram.
- **Visualization 2 (Continuous vs. Income):** Compare the distribution of a continuous variable (like `hours.per.week` or `age`) between the two income classes to observe distributional differences.
- **Visualization 3 (Custom Insight):** Create 1 custom visualization of your choice to uncover other interesting trends or relationships in the data.

Step 5 : Modeling and Prediction (Data Science)

- **Preparing Data for Modeling:** Convert all remaining categorical variables into "dummy" variables (one-hot encoding) so they can be used by the models.
- **Logistic Regression:** Train a logistic regression model to predict the income class. This is a good baseline model for understanding the influence of each variable.
- **Random Forest:** Train a Random Forest model, which generally performs better on this type of tabular data.
- **Evaluation and Interpretation:** Evaluate the models using a confusion matrix and the AUC score.
- **Feature Importance:** Extract the feature importances from the Random Forest model to identify the top 5 most important factors for predicting income.

Questions (answers will be put in the report or the notebook, as you want) :

1. Why is it dangerous to blindly drop all rows containing `"?"` in this specific dataset? How did consulting the metadata guide your imputation strategy?
2. Looking at your Unsupervised Learning results (PCA or K-Means), what hidden patterns did you discover that weren't immediately obvious from the raw data?
3. Comparing your Logistic Regression and Random Forest models, which metric (Precision, Recall, or AUC) do you consider most important for this specific business case, and why?
4. Based on your Random Forest feature importance, what actionable advice would you give to someone looking to maximize their chances of earning over $50K?